

The Department of Mining Engineering is located at the Takatu Campus, which is named after the Takatu Mountain in Quetta, Balochistan, and operates under the Faculty of Engineering and Architecture, a faculty established in 2004. The department itself was established in 2008, and since its inception, a total of eight batches have graduated and are currently serving across various public and private sector mining organizations. The department is fully accredited by the Pakistan Engineering Council and has a core focus on cultivating technical competence and applied research to resolve real-time issues of the mining industry. The curriculum focuses heavily on environmentally friendly mineral extraction, engineering professional ethics, workplace safety, and sustainable development. Its future ambitions include enhancing corporate and academic outreach by establishing sustainable linkages with local and international mining organizations. The mission of the Department of Mining Engineering is striving for a high-quality undergraduate program supported by a compatible contemporary curriculum, applied research, and the provision of technical competence to graduates possessing skills to contribute to the mining industry of Pakistan in general and Balochistan in particular with an environment-friendly approach. The Program Educational Objectives state that civil and mining engineering graduates are expected to demonstrate specific attributes three to five years post-graduation. These attributes include recognizing, investigating, and addressing engineering problems related to the exploitation of mineral deposits of Pakistan and Balochistan by employing modern techniques. Graduates are also expected to exhibit a high level of technical competence, applying research and problem-solving skills to produce solutions in the field of Mining Engineering. Furthermore, they must exercise professional excellence utilizing interpersonal skills focused on workplace safety, environmental sustainability, and professional engineering ethics, while pursuing lifelong learning by accepting practical challenges in the mining engineering field.

For the Bachelor of Science in Mining Engineering program, the admission requirements state that an applicant's academic background must include F.Sc Pre-Engineering or ICS with Mathematics and Physics from a recognized board or an equivalent qualification with at least sixty percent marks, or a Diploma of Associate Engineer in a relevant field with a minimum of sixty percent marks. The degree requirements mandate completing 136 total credit hours across 45 to 49 total courses, maintaining a minimum academic standing with a CGPA of 2.0 or higher, and mastering the twelve expected standard engineering graduate attributes such as Engineering Knowledge, Problem Analysis, and Modern Tool Usage. The curriculum plan outlines the program schema by semester, where credit hour structures denote theory plus lab hours. The first semester consists of seventeen total credit hours and includes Analytic Geometry and Calculus for three theory and zero lab hours, Applied Chemistry for three theory and one lab hours, Applied Physics for three theory and one lab hours, Engineering Drawing and Graphics for one theory and two lab hours, and Mining Engineering Fundamentals for three theory and zero lab hours. The second semester totals eighteen credit hours and covers Linear Algebra for three theory and zero lab hours, Applied Thermodynamics for three theory and one lab hours, Applied Geology for three theory and one lab hours, Islamic Studies or Ethics for two theory and zero lab hours, and English Composition and Comprehension or Professional Ethics for three or two theory and zero lab hours respectively. The third semester requires seventeen total credit hours, comprising Economic Minerals of Pakistan for three theory and zero lab hours, Engineering Mechanics for three theory and zero lab hours, Basic Electrical Technology for three theory and one lab hours, Fluid Mechanics for three theory and one lab hours, and Differential Equations for three theory and zero lab hours. The fourth semester encompasses eighteen total credit hours, including Communication Skills or Computer Programming and Lab for three or two theory

and zero or one lab hours respectively, Mine Surveying for three theory and one lab hours, Pakistan Studies for two theory and zero lab hours, Mechanics of Material for three theory and one lab hours, and Probability and Statistics for two theory and zero lab hours.

The fifth semester requires eighteen total credit hours and consists of Report Writing Skills or Numerical Method in Computing for two theory and zero lab hours each, Structural Geology for three theory and one lab hours, Mineral Exploration or Explosives Engineering for three theory and zero or one lab hours respectively, and Entrepreneurship for three theory and zero lab hours. The sixth semester involves seventeen total credit hours and includes Surface Mine Design for three theory and zero lab hours, Rock Mechanics for three theory and one lab hours, Mineral Processing-I for three theory and one lab hours, Mine Power, Drainage and Material Handling for three theory and zero lab hours, and Computer Applications in Mining Engineering for two theory and one lab hours. The seventh semester comprises eighteen total credit hours, featuring Mining Law or Mine Ventilation for two or three theory and zero or one lab hours respectively, Mineral Processing-II for three theory and one lab hours, Senior Design Project-I for zero theory and three lab hours, and Engineering Economics or Organizational Behavior for two or three theory and zero lab hours respectively. The eighth semester completes the undergraduate curriculum with sixteen total credit hours, requiring Senior Design Project-II for zero theory and three lab hours, Underground Mine Design or Mine Hazards and Safety for three theory and zero or one lab hours respectively, Engineering Management or Tunnel Engineering for two or three theory and zero or one lab hours respectively, and a First Aid Course for zero theory and zero lab hours. For the Master of Science Program, the admission requirements dictate an academic background of sixteen years of education or an equivalent benchmark qualification, such as a four-year BS or B.E. degree, in a relevant engineering discipline from an HEC-recognized university. Applicants must secure at least sixty percent marks under the annual grading setup or a minimum CGPA of 2.5 out of 4.0, though the raw web copy contains a localized placeholder text errantly listing Chemical or Petroleum fields. Candidates must also provide a valid GAT General score with a passing benchmark of at least fifty percent marks. The degree requirements for the MS program consist of thirty total credit hours, which include a total coursework requirement of eight advanced postgraduate courses, and students must maintain a minimum academic standing with a CGPA of 2.5 or higher.